

Censored linear regression models for irregularly observed longitudinal data using the multivariate- t distribution

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Abstract

In acquired immunodeficiency syndrome (AIDS) studies it is quite common to observe viral load measurements collected irregularly over time. Moreover, these measurements can be subjected to some upper and/or lower detection limits depending on the quantification assays. A complication arises when these continuous repeated measures have a heavy-tailed behavior. For such data structures, we propose a robust structure for a censored linear model based on the multivariate Student's t -distribution. To compensate for the autocorrelation existing among irregularly observed measures, a damped exponential correlation structure is employed. An efficient expectation maximization type algorithm is developed for computing the maximum likelihood estimates, obtaining as a by-product the standard errors of the fixed effects and the log-likelihood function. The proposed algorithm uses closed-form expressions at the E-step that rely on formulas for the mean and variance of a truncated multivariate Student's t -distribution. The methodology is illustrated through an application to an Human Immunodeficiency Virus-AIDS (HIV-AIDS) study and several simulation studies.

Keywords

censored data, expectation conditional maximization algorithm, longitudinal data, HIV viral load, outliers

1 Introduction

In many biomedical studies and clinical trials, the use of longitudinal models has shown a significant growth in recent years and becomes a powerful tool for modeling correlated outcomes. In clinical trials of antiretroviral therapy in acquired immunodeficiency syndrome (AIDS) studies, human immunodeficiency virus-1 (HIV-1) RNA (viral load) measures are collected over a period of

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treatment to determine rates of changes in the amount of actively replicating virus. These measures are used as a key primary endpoint because the viral load monitoring during the therapy is mostly available, a failure in the treatment can be defined virologically, and a new regimen of therapy is recommended as soon as virological rebound occurs.¹ Since for each patient the viral load measures are collected over time, the correlation structure among responses must be taken into account.

Longitudinal models allow us to estimate viral load trajectories as well as to quantify the correlation structure between viral load measurements.^{2,3} However, in practice, the statistical modeling of viral loads can be challenging due to the following features. First, the measurements can be subject to a upper/lower limit of quantification. As a result, the viral load responses are either left or right censored depending upon the diagnostics assay used. In general, the range of limit detection varies from 400 copies/ml for the earlier assays to 40 copies/ml for the more sophisticated assays of the recent times. Second, the viral loads are usually recorded at irregular occasions because the timings of measurements often vary from one patient to another, and typically measurement times are associated with the course of the disease. Third, the viral load measurements often contain influential and/or outlying observations, which can cause misleading results in parameter estimates as well as their standard errors when a Gaussian assumption is considered. Therefore, one of the greatest challenges related to longitudinal data modeling in AIDS research is to consider the inherent features of viral load measurements simultaneously.

In the statistical and biomedical literature, linear and nonlinear mixed effects models based on Gaussian assumptions are routinely used to model longitudinal data.⁴⁻⁶ Nevertheless, such an assumption could be not realistic because of the presence of atypical observations (outliers). To deal with this weakness, some alternatives based on heavy-tailed distributions have been proposed. For example, Pinheiro et al.⁷ proposed a Student's t linear mixed model (t -LMM) that has demonstrated its robustness against outliers. Other authors focused their research interest in developing strategies for fitting both linear and nonlinear mixed effects models under heavy-tailed distributions such as the Student's t , slash, and contaminated normal distributions.⁸⁻¹² The regression and mixed effects models for censored responses under heavy-tailed distributions have been studied in detail in the last years.¹³⁻¹⁶

Recently, a variety of heavy-tailed statistical models have been proposed for longitudinal data considering not only the correlation structure induced by the random effects term but also by other types of correlation in the error term. For example, Wang¹⁷ studied the multivariate t -LMM for outcome variables recorded at irregular occasions by using a parsimonious damping exponential correlation (DEC) structure. This type of correlation structure, proposed by Muñoz et al.,¹⁸ takes into account the autocorrelation generated by the within-subject dependence among irregular occasions. Wang and Fan¹⁹ considered the multivariate Student's t linear mixed with AR(p) dependence structure (a particular case of DEC structure) for the within-subject errors in the case of multiple outcomes. However, and as was mentioned by Goldstein et al.²⁰ and Browne and Goldstein,²¹ in cases when the repeated measures are collected close in time or correlations among the measures do not decay quickly, random effects modes may not adequately account for the dependency and a more complex correlation structure must be specified.

Following Wang,¹⁷ the aim of this paper is to consider the DEC structure for the across-occasion covariance matrix of the random errors under censored responses. As a consequence, the robust so-called Student's t multivariate linear censored (t -MLC) model with DEC structure is defined and a fully likelihood-based approach is conducted, including the implementation of an exact expectation conditional maximization (ECM) algorithm for the maximum likelihood (ML) estimation. As in Matos et al.,²² we show that the E-step reduces to computing the first two moments of certain truncated multivariate Student's t distribution, with the computation of the likelihood function and

the asymptotic standard errors as a by-product of the E-step. General formulas for these moments were derived by using the results given in Ho et al.²³ The likelihood function is used for monitoring convergence and the model selection is conducted through the Akaike information criterion (AIC) and Bayesian information criterion (BIC).

The rest of the paper is organized as follows. Section 2 describes a motivating real-life dataset of HIV-AIDS infected patients. In Section 3 we introduce some notation related to the truncated Student's t -distribution. Then, the t -MLC is presented. In Section 4, the related likelihood-based inferences are presented including the imputation procedure of censored cases. The method for the prediction of future observations is presented in Section 5. Section 6 presents two simulation studies for comparing the performance of our method with other normality-based one. The advantage of the proposed method is presented through the analysis of case studies of HIV viral load in Section 7. Section 8 concludes with a short discussion of issues raised by our study and some possible directions for a future research.

2 Motivating example: Unstructured treatment interruption (UTI) data

In this section, we present a longitudinal dataset corresponding to UTI—unstructured antiretroviral therapy treatment interruption—in HIV-infected adolescents in four institutions in the US. In this case, the HIV-1 RNA measures are subject to censoring below the lower limit of detection of the assay (50 copies/ml) and the presence of outlying and influential observations is noted.

This dataset consist on the measurements of HIV-1 RNA measures after unstructured treatment interruption in 72 adolescents from US. UTI was defined as discontinuation of all antiretroviral drugs for any period of time, after which treatment was resumed. The reasons for interruption might have been diverse. For example, Saitoh et al.²⁴ mentioned (a) the medication fatigue, (b) patients inability to take antiretroviral medications, (c) toxicity associated with the use of antiretroviral medications, and (d) adverse effects.

The dataset presents about 7% of observations below the detection limits of assay quantifications (left censored). The viral loads were monitored from the closest time points at 0, 1, 3, 6, 9, 12, 18, and 24 months after the treatment interruption, that is, they were irregularly collected over time. A more detailed explanation of the data can be found in Saitoh et al.²⁴ and Vaida and Liu.²⁵ The individual profiles of viral load at different follow-up times after UTI appear in Figure 1 (panel a). This figure also presents the normal quantile–quantile (Q–Q) plot for the residuals (panel b) obtained by fitting a censored (Gaussian) mixed effect model proposed by Vaida and Liu²⁵ using the *lmec* package of R. The Q–Q plot exhibits a heavy-tailed behavior, suggesting that the normality assumption for the within-subject errors might be inappropriate. The nonnormality of the distribution gives an indication that some atypical observations or outliers might exist in the data. Outliers may cause misleading results in parameter estimates as well as their standard errors and they could have an enormous impact on statistical inferences. In addition, Table 1 presents the observed correlation for the responses in different time points. Clearly, the data do not appear to satisfy an uncorrelated structure of zero covariances or a compound symmetry assumption of equal variances and covariances across time.

3 Model specification

3.1 Preliminaries

In this section, we present some useful results associated to the p -variate Student's t -distribution that will be needed for implementing the expectation maximization (EM) algorithm. We start with the

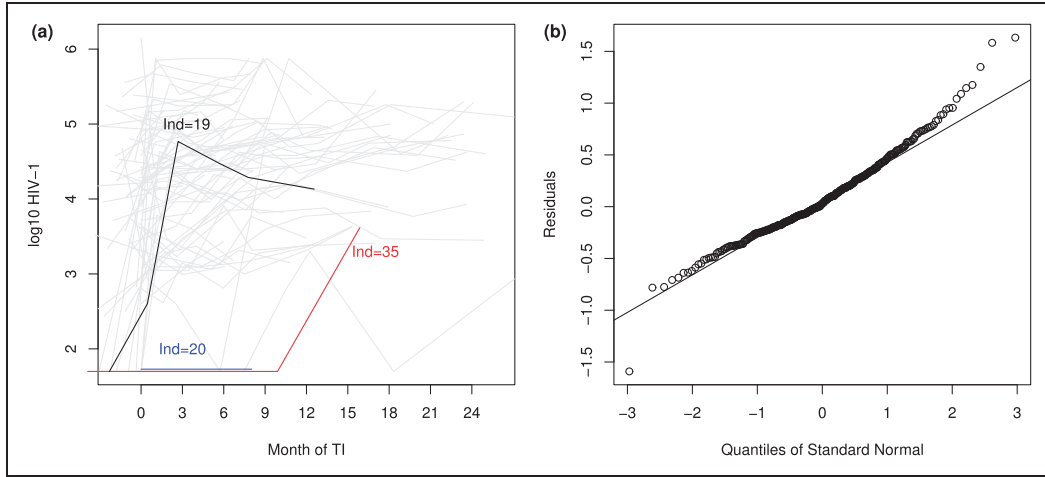


Figure 1. UTI data. (a) Individual profiles (in $\log_{10}$ scale) for HIV viral load at different follow-up times. Trajectories for three censored individuals are indicated in different colors. The abbreviation “Ind” stands for “Individual.” (b) Normal Q–Q plot for model residuals obtained by using the *lme4* package of R.

Table 1. Observed correlation of $\log_{10}$ RNA for a single response over different times.

		$\log_{10}$ RNA							
		Month 0	Month 1	Month 3	Month 6	Month 9	Month 12	Month 18	Month 24
$\log_{10}$ RNA	Month 0		0.4877	0.4100	0.4052	0.4820	0.4435	0.3441	0.6529
	Month 1	0.4877		0.9145	0.8551	0.8455	0.6978	0.7090	0.6140
	Month 3	0.4100	0.9145		0.9255	0.8638	0.7209	0.7601	0.6301
	Month 6	0.4052	0.8551	0.9255		0.8238	0.6490	0.6548	0.5314
	Month 9	0.4820	0.8455	0.8638	0.8238		0.9185	0.7642	0.8061
	Month 12	0.4435	0.6978	0.7209	0.6490	0.9185		0.6646	0.6897
	Month 18	0.3441	0.7090	0.7601	0.6548	0.7642	0.6646		0.8947
	Month 24	0.6529	0.6140	0.6301	0.5314	0.8061	0.6897	0.8947	

probability density function (*pdf*) of a Student’s *t*-distribution random vector $\mathbf{Y} \in \mathbb{R}^p$ with location vector $\boldsymbol{\mu}$, scale matrix $\boldsymbol{\Sigma}$, and ν degrees of freedom. Its *pdf* is given by

$$t_p(\mathbf{y}|\boldsymbol{\mu}, \boldsymbol{\Sigma}, \nu) = \frac{\Gamma(\frac{p+\nu}{2})}{\Gamma(\frac{\nu}{2})\pi^{p/2}} \nu^{-p/2} |\boldsymbol{\Sigma}|^{-1/2} \left(1 + \frac{\delta}{\nu}\right)^{-(p+\nu)/2}$$

where $\Gamma(\cdot)$ is the standard gamma function and $\delta = (\mathbf{y} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{y} - \boldsymbol{\mu})$ is the Mahalanobis distance. The notation adopted for the Student’s *t pdf* is $t_p(\boldsymbol{\mu}, \boldsymbol{\Sigma}, \nu)$.

The cumulative distribution function (*cdf*) is denoted by $T_p(\cdot|\boldsymbol{\mu}, \boldsymbol{\Sigma}, \nu)$. It is important to stress that if $\nu > 1$, the mean of $\mathbf{y}$ is $\boldsymbol{\mu}$ and if $\nu > 2$, the covariance matrix is given by $\nu(\nu - 2)^{-1}\boldsymbol{\Sigma}$. Moreover, as ν tends to infinity, $\mathbf{Y}$ converges in distribution to a multivariate normal with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$.

An important property of the random vector $\mathbf{Y}$ is that it can be written as a mixture of a normal random vector and a positive random variable, i.e.

$$\mathbf{Y} = \boldsymbol{\mu} + U^{-1/2} \mathbf{Z}$$

where $\mathbf{Z}$ is a normal random vector, with zero-mean vector and covariance $\boldsymbol{\Sigma}$, independent of U , which is a positive random variable with a gamma distribution $\text{Gamma}(\nu/2, \nu/2)$.¹

The distribution of $\mathbf{Y}$ constrained to lie within the right-truncated hyperplane

$$\mathbb{A} = \{\mathbf{y} \in \mathbb{R}^p | \mathbf{y} \leq \mathbf{a}\} \quad (1)$$

where $\mathbf{y} = (y_1, \dots, y_p)^\top$ and $\mathbf{a} = (a_1, \dots, a_p)^\top$ is a truncated Student's t -distribution, denoted by $Tt_p(\boldsymbol{\mu}, \boldsymbol{\Sigma}, \nu; \mathbb{A})$, with *pdf* given by $f(\mathbf{y} | \boldsymbol{\mu}, \boldsymbol{\Sigma}, \nu; \mathbb{A}) = \frac{t_p(\mathbf{y} | \boldsymbol{\mu}, \boldsymbol{\Sigma}, \nu)}{T_p(\mathbf{a} | \boldsymbol{\mu}, \boldsymbol{\Sigma}, \nu)} \mathbb{I}_{\mathbb{A}}(\mathbf{y})$, where $\mathbb{I}_{\mathbb{A}}(\mathbf{y})$ is the indicator function of $\mathbb{A}$.

As was mentioned at the beginning of this section, the following properties of the multivariate Student's t and truncated Student's t -distribution are useful for the implementation of the EM algorithm. We start with the marginal-conditional decomposition of a Student's t random vector. Details of the proofs are provided in Arellano-Valle and Bolfarine.²⁶

Proposition 1: Let $\mathbf{Y} \sim t_p(\boldsymbol{\mu}, \boldsymbol{\Sigma}, \nu)$ and $\mathbf{Y}$ be partitioned as $\mathbf{Y}^\top = (\mathbf{Y}_1^\top, \mathbf{Y}_2^\top)^\top$, with $\dim(\mathbf{Y}_1) = p_1$, $\dim(\mathbf{Y}_2) = p_2$, $p_1 + p_2 = p$, and where $\boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{pmatrix}$ and $\boldsymbol{\mu} = (\boldsymbol{\mu}_1^\top, \boldsymbol{\mu}_2^\top)^\top$, are the corresponding partitions of $\boldsymbol{\Sigma}$ and $\boldsymbol{\mu}$. Then, we have

- (i) $\mathbf{Y}_1 \sim t_{p_1}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_{11}, \nu)$; and
- (ii) the conditional *cdf* of $\mathbf{Y}_2 | \mathbf{Y}_1 = \mathbf{y}_1$ is given by

$$P(\mathbf{Y}_2 \leq \mathbf{y}_2 | \mathbf{Y}_1 = \mathbf{y}_1) = T_{p_2}(\mathbf{y}_2 | \boldsymbol{\mu}_{2.1}, \tilde{\boldsymbol{\Sigma}}_{22.1}, \nu + p_1)$$

where

$$\begin{aligned} \tilde{\boldsymbol{\Sigma}}_{22.1} &= \left(\frac{\nu + \delta_1}{\nu + p_1} \right) \boldsymbol{\Sigma}_{22.1}, \quad \delta_1 = (\mathbf{y}_1 - \boldsymbol{\mu}_1)^\top \boldsymbol{\Sigma}_{11}^{-1} (\mathbf{y}_1 - \boldsymbol{\mu}_1), \quad \boldsymbol{\Sigma}_{22.1} = \boldsymbol{\Sigma}_{22} - \boldsymbol{\Sigma}_{21} \boldsymbol{\Sigma}_{11}^{-1} \boldsymbol{\Sigma}_{12}, \quad \text{and} \\ \boldsymbol{\mu}_{2.1} &= \boldsymbol{\mu}_2 + \boldsymbol{\Sigma}_{21} \boldsymbol{\Sigma}_{11}^{-1} (\mathbf{y}_1 - \boldsymbol{\mu}_1). \end{aligned}$$

The following results provide the truncated moments of a Student's t random vector. The proofs of Proposition 2 and 3 are given in Matos et al.²² The proof of Proposition 4 is given in Ho et al.²³

Proposition 2: If $\mathbf{Y} \sim Tt_p(\boldsymbol{\mu}, \boldsymbol{\Sigma}, \nu; \mathbb{A})$ with $\mathbb{A}$ as (1), then the k -th moment of $\mathbf{Y}$, $k = 0, 1, 2$, is

$$E \left[\left(\frac{\nu + p}{\nu + \delta} \right)^r \mathbf{Y}^{(k)} \right] = c_p(\nu, r) \frac{T_p(\mathbf{a} | \boldsymbol{\mu}, \boldsymbol{\Sigma}^*, \nu + 2r)}{T_p(\mathbf{a} | \boldsymbol{\mu}, \boldsymbol{\Sigma}, \nu)} E_{\mathbf{W}}[\mathbf{W}^{(k)}], \quad \mathbf{W} \sim Tt_p(\boldsymbol{\mu}, \boldsymbol{\Sigma}^*, \nu + 2r; \mathbb{A})$$

where

$$c_p(v, r) = \left(\frac{v+p}{v}\right)^r \left(\frac{\Gamma((p+v)/2)\Gamma((v+2r)/2)}{\Gamma(v/2)\Gamma((p+v+2r)/2)}\right), \quad \delta = (\mathbf{Y} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{Y} - \boldsymbol{\mu}), \quad \mathbf{a} = (a_1, \dots, a_p)^\top, \\ \boldsymbol{\Sigma}^* = \frac{v}{v+2r} \boldsymbol{\Sigma}, \quad \mathbf{Y}^{(0)} = 1, \quad \mathbf{Y}^{(1)} = \mathbf{Y}, \quad \mathbf{Y}^{(2)} = \mathbf{Y}\mathbf{Y}^\top \quad \text{and} \quad v+2r > 0.$$

Having established a formula for the k -order moments of $\mathbf{Y}$, we now present a result on conditional moments of the partition of $\mathbf{Y}$.

Proposition 3: Let $\mathbf{Y} \sim Tt_p(\boldsymbol{\mu}, \boldsymbol{\Sigma}, v; \mathbb{A})$ with $\mathbb{A}$ as (I). Consider the partition $\mathbf{Y}^\top = (\mathbf{Y}_1^\top, \mathbf{Y}_2^\top)$ with $\dim(\mathbf{Y}_1) = p_1$, $\dim(\mathbf{Y}_2) = p_2$, $p_1 + p_2 = p$, and the corresponding partition of the parameters $\boldsymbol{\mu}, \boldsymbol{\Sigma}, \mathbf{a}(\mathbf{a}^{y_1}, \mathbf{a}^{y_2})$ and $\mathbb{A}(\mathbb{A}^{y_1}, \mathbb{A}^{y_2})$. Then, under the notation of Proposition 1, the conditional k -th moment of $\mathbf{Y}_2$ is

$$E\left[\left(\frac{v+p}{v+\delta}\right)^r \mathbf{Y}_2^{(k)} | \mathbf{Y}_1\right] = \frac{d_p(p_1, v, r)}{(v+\delta_1)^r} \frac{T_{p_2}(\mathbf{a}^{y_2} | \boldsymbol{\mu}_{2,1}, \tilde{\boldsymbol{\Sigma}}_{22,1}^*, v+p_1+2r)}{T_{p_2}(\mathbf{a}^{y_2} | \boldsymbol{\mu}_{2,1}, \tilde{\boldsymbol{\Sigma}}_{22,1}, v+p_1)} E_{\mathbf{W}}[\mathbf{W}^{(k)}],$$

where

$$\mathbf{W} \sim Tt_{p_2}(\boldsymbol{\mu}_{2,1}, \tilde{\boldsymbol{\Sigma}}_{22,1}^*, v+p_1+2r; \mathbb{A}^{y_2}), \quad \delta = (\mathbf{Y} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{Y} - \boldsymbol{\mu}), \\ \delta_1 = (\mathbf{Y}_1 - \boldsymbol{\mu}_1)^\top \boldsymbol{\Sigma}_{11}^{-1}(\mathbf{Y}_1 - \boldsymbol{\mu}_1), \quad \mathbf{a}^{y_2} = (a_1, \dots, a_{p_2})^\top, \quad \tilde{\boldsymbol{\Sigma}}_{22,1}^* = \left(\frac{v+\delta_1}{v+2r+p_1}\right) \boldsymbol{\Sigma}_{22,1}, \\ v+p_1+2r > 0 \quad \text{and} \quad d_p(p_1, v, r) = (v+p)^r \left(\frac{\Gamma((p+v)/2)\Gamma((p_1+v+2r)/2)}{\Gamma((p_1+v)/2)\Gamma((p+v+2r)/2)}\right).$$

In the following proposition, we establish relationships between the expectation and covariance of $\mathbf{Y}$ and $\mathbf{W}$.

Proposition 4: Let $\mathbf{Y} \sim Tt_p(\boldsymbol{\mu}, \boldsymbol{\Sigma}, v; \mathbb{A}^*)$, with $\mathbb{A}^* = \{\mathbf{y} \in \mathbb{R}^p | \mathbf{a}^* < \mathbf{y} \leq \mathbf{b}^*\}$, where $\mathbf{a}^* = (a_1^*, \dots, a_p^*)^\top$, $\mathbf{b}^* = (b_1^*, \dots, b_p^*)^\top$, $\boldsymbol{\Sigma} = \Lambda \mathbf{R} \Lambda$ and $\Lambda = \text{Diag}(\sigma_{11}, \dots, \sigma_{pp})$ is a $p \times p$ diagonal matrix with each element being positive. We have that $\mathbf{W} = \Lambda^{-1}(\mathbf{Y} - \boldsymbol{\mu}) \sim Tt_p(\mathbf{0}, \mathbf{R}, v; \mathbb{A})$ where $\mathbf{a} = \Lambda^{-1}(\mathbf{a}^* - \boldsymbol{\mu})$ and $\mathbf{b} = \Lambda^{-1}(\mathbf{b}^* - \boldsymbol{\mu})$. Therefore

$$E[\mathbf{Y}] = \boldsymbol{\mu} + \Lambda E[\mathbf{W}] \\ E[\mathbf{Y}\mathbf{Y}^\top] = \boldsymbol{\mu}\boldsymbol{\mu}^\top + \Lambda E[\mathbf{W}]\boldsymbol{\mu}^\top + \boldsymbol{\mu}E[\mathbf{W}^\top]\Lambda + \Lambda E[\mathbf{W}\mathbf{W}^\top]\Lambda^\top$$

where $E[\mathbf{W}]$ and $E[\mathbf{W}\mathbf{W}^\top]$ are given in Ho et al.²³

3.2 The statistical model

Our multivariate Student's t linear model, for longitudinal responses, is defined by

$$\mathbf{Y}_i = \mathbf{X}_i \boldsymbol{\beta} + \boldsymbol{\epsilon}_i \quad (2)$$

with $\epsilon_i \sim t_{n_i}\{\mathbf{0}, \Sigma_i, \nu\}$, where $\mathbf{Y}_i = (Y_{i1}, \dots, Y_{in_i})^\top$ is a $n_i \times 1$ vector of continuous responses for sample unit i measured at particular time points $\mathbf{t}_i = (t_{i1}, \dots, t_{in_i})^\top$, $\mathbf{X}_i$ is the $n_i \times p$ design matrix corresponding to the $p \times 1$ vector of fixed effects β , and ϵ_i is the $n_i \times 1$ vector of random errors. As was noted in Section 2, the measurements of the viral load for each subject present evidence of serial correlation. Therefore, to obtain accurate parameter estimates, we consider a parsimonious structure on the dispersion matrix $\Sigma_i = \sigma^2 \mathbf{E}_i$, where the matrix $\mathbf{E}_i$ incorporates a time-dependence structure. Thus, we adopt the DEC structure for Σ_i , as proposed by Muñoz et al.¹⁸ This correlation structure allows us to deal with unequally spaced and unbalanced observations and is defined as

$$\Sigma_i = \sigma^2 \mathbf{E}_i = \sigma^2 \mathbf{E}_i(\phi, \mathbf{t}_i) = \sigma^2 \left[\phi_1^{|t_{ij} - t_{ik}|^{\phi_2}} \right] \quad (3)$$

for $i = 1, \dots, n, j, k = 1, \dots, n_i$. The correlation parameter ϕ_1 describes the autocorrelation between observations separated by the absolute length of two time points, and the damping parameter ϕ_2 allows the acceleration of the exponential decay of the autocorrelation function defining a continuous-time autoregressive (AR) model. It is important to stress that considering the DEC structure it is possible to obtain different correlation structures. For example, for a given nonnegative ϕ_1 :

- If $\phi_2 = 0$, then the $\mathbf{E}_i$ is the compound symmetry structure.
- If $0 < \phi_2 < 1$, then we have $\mathbf{E}_i$ with a decay rate between that of compound symmetry and the first-order AR (AR(1)) model.
- If $\phi_2 = 1$, then the $\mathbf{E}_i$ is an AR(1) model.
- If $\phi_2 > 1$, then we have $\mathbf{E}_i$ with a decay rate faster than AR(1) model.
- If $\phi_2 \rightarrow \infty$, then the $\mathbf{E}_i$ is the first-order moving average (MA(1)) model.

For a more detailed discussion about the DEC structure, we refer to Muñoz et al.¹⁸ and Wang.¹⁷

From a practical point of view and in order to avoid computational problems, the parameter space of ϕ_1 and ϕ_2 is confined within $\{(\phi_1, \phi_2) : 0 < \phi_1 < 1, \phi_2 > 0\}$, that is, we restrict our attention to nonnegative values of ϕ_1 and ϕ_2 . Such assumption is quite common in most biomedical and epidemiological applications. Finally, we consider the approach proposed by Vaida and Liu²⁵ and Matos et al.²² to modeling the censored responses. Thus, the observed data for the i th subject are given by $(\mathbf{V}_i, \mathbf{C}_i)$, where $\mathbf{V}_i$ represents the vector of uncensored reading or censoring level and $\mathbf{C}_i$ the vector of censoring indicators. In other words

$$y_{ij} \leq V_{ij} \quad \text{if } C_{ij} = 1, \quad \text{and} \quad y_{ij} = V_{ij} \quad \text{if } C_{ij} = 0 \quad (4)$$

so that, considering equation (4) along with equations (2) to (3) defined the multivariate Student's t linear censored (t -MLC) model. Notice that a left censoring structure causes a right truncation of the distribution, since we only know that the true observation y_{ij} is less than or equal to the observed quantity V_{ij} . Moreover, the right censored problem can be represented by a left censored problem by simultaneously transforming the response y_{ij} and censoring level V_{ij} to $-y_{ij}$ and $-V_{ij}$.

3.3 The likelihood function

To obtain the likelihood function of the t -MLC model, first we treat separately the observed and censored components of $\mathbf{y}_i$, i.e. $\mathbf{y}_i = (\mathbf{y}_i^o, \mathbf{y}_i^c)^\top$, with $C_{ij}=0$ for all elements in $\mathbf{y}_i^o$, and $C_{ij}=1$ for all elements in $\mathbf{y}_i^c$. Analogous, we write $\mathbf{V}_i = \text{vec}(\mathbf{V}_i^o, \mathbf{V}_i^c)$, where $\text{vec}(\cdot)$ denotes the function which stacks

vectors or matrices of the same number of columns, with $\Sigma_i = \begin{pmatrix} \Sigma_i^{oo} & \Sigma_i^{oc} \\ \Sigma_i^{co} & \Sigma_i^{cc} \end{pmatrix}$. Then, using Proposition 1, we have that $\mathbf{y}_i^o \sim t_{n_i^o}(\mathbf{X}_i^o \boldsymbol{\beta}, \Sigma_i^{oo}, \nu)$ and $\mathbf{y}_i^c | \mathbf{y}_i^o \sim t_{n_i^c}(\boldsymbol{\mu}_i^{co}, \mathbf{S}_i^{co}, \nu + n_i^o)$, where

$$\boldsymbol{\mu}_i^{co} = \mathbf{X}_i^c \boldsymbol{\beta} + \Sigma_i^{co} \Sigma_i^{oo-1} (\mathbf{y}_i^o - \mathbf{X}_i^o \boldsymbol{\beta}), \quad \mathbf{S}_i^{co} = \left(\frac{\nu + Q(\mathbf{y}_i^o)}{\nu + n_i^o} \right) \Sigma_i^{cc.o} \quad (5)$$

with $\Sigma_i^{cc.o} = \Sigma_i^{cc} - \Sigma_i^{co} \Sigma_i^{oo-1} \Sigma_i^{oc}$ and $Q(\mathbf{y}_i^o) = (\mathbf{y}_i^o - \mathbf{X}_i^o \boldsymbol{\beta})^\top \Sigma_i^{oo-1} (\mathbf{y}_i^o - \mathbf{X}_i^o \boldsymbol{\beta})$. Therefore, the likelihood function for subject i is

$$\begin{aligned} L_i(\boldsymbol{\theta} | \mathbf{y}) &= f(\mathbf{V}_i | \mathbf{C}_i, \boldsymbol{\theta}) = f(\mathbf{y}_i^c \leq \mathbf{V}_i^c | \mathbf{y}_i^o = \mathbf{V}_i^o, \boldsymbol{\theta}) f(\mathbf{y}_i^o = \mathbf{V}_i^o | \boldsymbol{\theta}), \\ &= T_{n_i^c}(\mathbf{V}_i^c | \boldsymbol{\mu}_i^{co}, \mathbf{S}_i^{co}, \nu + n_i^o) t_{n_i^o}(\mathbf{V}_i^o | \mathbf{X}_i^o \boldsymbol{\beta}, \Sigma_i^{oo}, \nu) = L_i \end{aligned}$$

Straightforwardly, the log-likelihood function for the observed data is given by $\ell(\boldsymbol{\theta} | \mathbf{y}) = \sum_{i=1}^n \log L_i$. It is important to note that this function can be computed at each step of the EM-type algorithm without additional computational burden since the L_i 's have already been computed at the E-step. We assume that the degrees of freedom parameter of the Student's t distribution is fixed. For choosing the most appropriate value of this parameter, we will use the log-likelihood profile.^{10,27} This assumption is based on the work by Lucas,²⁸ in which the author showed that the protection against outliers is preserved only if the degrees of freedom parameter is fixed. Consequently, the parameter vector for the t -MLC model is $\boldsymbol{\theta} = (\boldsymbol{\beta}^\top, \sigma^2, \phi_1, \phi_2)^\top$.

4 The EM algorithm

We describe in detail how to carry out ML estimation for the proposed t -MLC model. The EM algorithm, originally proposed by Dempster et al.,²⁹ is a very popular iterative optimization strategy commonly used to obtain ML estimates for incomplete data problems. This algorithm has many attractive features such as the numerical stability and the simplicity of implementation and its memory requirements are quite reasonable.³⁰ However, ML estimation for the t -MLC model is complicated because of the censoring and the DEC structure, and the EM algorithm is less advisable due to the computational difficulty at the M-step. To overcome this problem, we used an extension of the EM algorithm, called the ECM algorithm.³¹ A key feature of this algorithm is that it preserves the stability of the EM and has a typically faster convergence rate than the original EM.

In order to propose the ECM algorithm for our t -MLC model, firstly we define $\mathbf{y} = (\mathbf{y}_1^\top, \dots, \mathbf{y}_n^\top)^\top$, $\mathbf{u} = (u_1, \dots, u_n)^\top$, $\mathbf{V} = \text{vec}(\mathbf{V}_1, \dots, \mathbf{V}_n)$, and $\mathbf{C} = \text{vec}(\mathbf{C}_1, \dots, \mathbf{C}_n)$ such that we observe $(\mathbf{V}_i, \mathbf{C}_i)$ for the i -th subject. Now, we treat $\mathbf{u}$ and $\mathbf{y}$ as hypothetical missing data, and augmenting with the observed data $\mathbf{V}$, $\mathbf{C}$ corresponding to the censoring mechanism. Consequently, we set the complete data vector as $\mathbf{y}_c = (\mathbf{C}^\top, \mathbf{V}^\top, \mathbf{y}^\top, \mathbf{u}^\top)^\top$. As is well known, the ECM algorithm must be applied to the complete data log-likelihood function given by

$$\ell_c(\boldsymbol{\theta} | \mathbf{y}_c) = \sum_{i=1}^n \ell_i(\boldsymbol{\theta} | \mathbf{y}_c)$$

where

$$\ell_i(\boldsymbol{\theta} | \mathbf{y}_c) = -\frac{1}{2} \left[n_i \log \sigma^2 + \log |\mathbf{E}_i| + \frac{u_i}{\sigma^2} (\mathbf{y}_i - \mathbf{X}_i \boldsymbol{\beta})^\top \mathbf{E}_i^{-1} (\mathbf{y}_i - \mathbf{X}_i \boldsymbol{\beta}) \right] + h(u_i | \nu) + c$$

c is a constant that does not depend on θ and $h(u_i|v)$ is the Gamma($v/2, v/2$) *pdf*. Finally, the ECM algorithm for the t -MLC model can be summarized through the following two steps.

E-step:

Given the current value $\theta = \widehat{\theta}^{(k)}$, the E-step provides the conditional expectation of the complete data log-likelihood function

$$Q(\theta|\widehat{\theta}^{(k)}) = \sum_{i=1}^n Q_i(\theta|\widehat{\theta}^{(k)}) \quad (6)$$

where

$$Q_i(\beta, \sigma^2, \phi|\widehat{\theta}^{(k)}) = -\frac{n_i}{2} \log \sigma^2 - \frac{1}{2} \log |\mathbf{E}_i| - \frac{1}{2\sigma^2} A_i^{(k)}(\beta, \phi)$$

with

$$A_i^{(k)}(\beta, \phi) = \left[\text{tr}(u\widehat{\mathbf{y}}_i^{2(k)} \mathbf{E}_i^{-1}) - 2\beta \mathbf{X}_i^\top \mathbf{E}_i^{-1} u\widehat{\mathbf{y}}_i^{(k)} + \widehat{u}_i^{(k)} \beta \mathbf{X}_i^\top \mathbf{E}_i^{-1} \mathbf{X}_i \beta \right]$$

Note that, since v is fixed, there is no need to obtain $E[h(u_i|v)|\mathbf{V}, \mathbf{C}, \widehat{\theta}^{(k)}]$.

Conditional maximization (CM) step:

In this step, $Q(\theta|\widehat{\theta}^{(k)})$ is conditionally maximized with respect to θ and a new estimate $\widehat{\theta}^{(k+1)}$ is obtained. Specifically, we have that

$$\widehat{\beta}^{(k+1)} = \left(\sum_{i=1}^n \widehat{u}_i^{(k)} \mathbf{X}_i^\top (\widehat{\mathbf{E}}_i^{(k)})^{-1} \mathbf{X}_i \right)^{-1} \sum_{i=1}^n \mathbf{X}_i^\top (\widehat{\mathbf{E}}_i^{(k)})^{-1} u\widehat{\mathbf{y}}_i^{(k)} \quad (7)$$

$$\widehat{\sigma}^{2(k+1)} = \frac{1}{N} \sum_{i=1}^n A_i^{(k)}(\widehat{\beta}^{(k+1)}, \widehat{\phi}^{(k)}) \quad (8)$$

$$\widehat{\phi}^{(k+1)} = \underset{\phi}{\text{argmax}} \left\{ -\frac{1}{2} \sum_{i=1}^n \left[\log(|\mathbf{E}_i|) + A_i^{(k)}(\widehat{\beta}^{(k+1)}, \phi) \right] \right\} \quad (9)$$

where $N = \sum_{i=1}^n n_i$. The *optim* routine in R package³² can be used to perform a two-dimensional search of (ϕ_1, ϕ_2) in equation (9) subject to box constraints: $\phi_1 \in [0, 1)$ and $\phi_2 \in (0, \infty)$.

The algorithm is iterated until a suitable convergence rule is satisfied. In this case, we adopt the distance involving two successive evaluations of the log-likelihood $|\ell(\widehat{\theta}^{(k+1)})/\ell(\widehat{\theta}^{(k)}) - 1|$ as a convergence criterion. It is important to stress that from equations (7) to (9), the E-step reduces to the computation of $u\widehat{\mathbf{y}}_i^2$, $u\widehat{\mathbf{y}}_i$, and $\widehat{u}_i$. These expected values can be determined in closed form, using Propositions 1–4, as follows:

(1) If the subject i has only censored components, from Proposition 2

$$u\widehat{\mathbf{y}}_i^2 = E[u_i \mathbf{y}_i \mathbf{y}_i^\top | \mathbf{V}_i, \mathbf{C}_i, \widehat{\theta}^{(k)}] = \frac{T_{n_i}(\mathbf{V}_i | \widehat{\boldsymbol{\mu}}_i^{(k)}, \widehat{\boldsymbol{\Sigma}}_i^{*(k)}, v+2)}{T_{n_i}(\mathbf{V}_i | \widehat{\boldsymbol{\mu}}_i^{(k)}, \widehat{\boldsymbol{\Sigma}}_i^{(k)}, v)} E[\mathbf{W}_i \mathbf{W}_i^\top],$$

$$\widehat{u}\mathbf{y}_i = E[u_i \mathbf{y}_i | \mathbf{V}_i, \mathbf{C}_i, \widehat{\boldsymbol{\theta}}^{(k)}] = \frac{T_{n_i}(\mathbf{V}_i | \widehat{\boldsymbol{\mu}}_i^{(k)}, \widehat{\boldsymbol{\Sigma}}_i^{*(k)}, \nu + 2)}{T_{n_i}(\mathbf{V}_i | \widehat{\boldsymbol{\mu}}_i^{(k)}, \widehat{\boldsymbol{\Sigma}}_i^{(k)}, \nu)} E[\mathbf{W}_i],$$

$$\widehat{u}_i = E[u_i | \mathbf{V}_i, \mathbf{C}_i, \widehat{\boldsymbol{\theta}}^{(k)}] = \frac{T_{n_i}(\mathbf{V}_i | \widehat{\boldsymbol{\mu}}_i^{(k)}, \widehat{\boldsymbol{\Sigma}}_i^{*(k)}, \nu + 2)}{T_{n_i}(\mathbf{V}_i | \widehat{\boldsymbol{\mu}}_i^{(k)}, \widehat{\boldsymbol{\Sigma}}_i^{(k)}, \nu)}$$

where $\mathbf{W}_i \sim T_{n_i}(\widehat{\boldsymbol{\mu}}_i^{(k)}, \widehat{\boldsymbol{\Sigma}}_i^{*(k)}, \nu + 2; \mathbb{A}_i)$, $\widehat{\boldsymbol{\mu}}_i^{(k)} = \mathbf{X}_i \widehat{\boldsymbol{\beta}}^{(k)}$, $\widehat{\boldsymbol{\Sigma}}_i^{*(k)} = \frac{\nu}{\nu+2} \widehat{\boldsymbol{\Sigma}}_i^{(k)}$, $\widehat{\boldsymbol{\Sigma}}_i^{(k)} = \sigma^{2(k)} \widehat{\mathbf{E}}_i^{(k)}$, and $\mathbb{A}_i = \{\mathbf{W}_i \in \mathbb{R}^{n_i} | \mathbf{w}_i \leq \mathbf{V}_i\}$ where $\mathbf{w}_i = (w_{i1}, \dots, w_{ini})^\top$ and $\mathbf{V}_i = (V_{i1}, \dots, V_{ini})^\top$.

(2) If the subject i has only noncensored components, then

$$\widehat{u}\mathbf{y}_i^2 = \left(\frac{\nu + n_i}{\nu + Q(\mathbf{y}_i)} \right) \mathbf{y}_i \mathbf{y}_i^\top, \quad \widehat{u}\mathbf{y}_i = \left(\frac{\nu + n_i}{\nu + Q(\mathbf{y}_i)} \right) \mathbf{y}_i, \quad \widehat{u}_i = \left(\frac{\nu + n_i}{\nu + Q(\mathbf{y}_i)} \right)$$

where $Q(\mathbf{y}_i) = (\mathbf{y}_i - \mathbf{X}_i \boldsymbol{\beta})^\top \boldsymbol{\Sigma}_i^{-1} (\mathbf{y}_i - \mathbf{X}_i \boldsymbol{\beta})$.

(3) If the subject i has censored and uncensored components, then from Proposition 3 and given that $\{\mathbf{y}_i | \mathbf{V}_i, \mathbf{C}_i\}$, $\{\mathbf{y}_i^c | \mathbf{V}_i, \mathbf{C}_i, \mathbf{y}_i^o\}$, and $\{\mathbf{y}_i^c | \mathbf{V}_i, \mathbf{C}_i, \mathbf{y}_i^o\}$ are equivalent processes, we have that

$$\widehat{u}\mathbf{y}_i^2 = E[u_i \mathbf{y}_i \mathbf{y}_i^\top | \mathbf{y}_i^o, \mathbf{V}_i, \mathbf{C}_i, \widehat{\boldsymbol{\theta}}^{(k)}] = \begin{pmatrix} \mathbf{y}_i^o \mathbf{y}_i^{o\top} \widehat{u}_i & \widehat{u}_i \mathbf{y}_i^o \widehat{\mathbf{w}}_i^{c\top} \\ \widehat{u}_i \widehat{\mathbf{w}}_i^c \mathbf{y}_i^{o\top} & \widehat{u}_i \widehat{\mathbf{w}}_i^{2c} \end{pmatrix},$$

$$\widehat{u}\mathbf{y}_i = E[u_i \mathbf{y}_i | \mathbf{y}_i^o, \mathbf{V}_i, \mathbf{C}_i, \widehat{\boldsymbol{\theta}}^{(k)}] = \text{vec}(\mathbf{y}_i^o \widehat{u}_i, \widehat{\mathbf{w}}_i^c),$$

$$\widehat{u}_i = E[u_i | \mathbf{y}_i^o, \mathbf{V}_i, \mathbf{C}_i, \widehat{\boldsymbol{\theta}}^{(k)}] = \left(\frac{n_i^o + \nu}{\nu + Q(\mathbf{y}_i^o)} \right) \frac{T_{n_i}(\mathbf{V}_i | \boldsymbol{\mu}_i^{co}, \widetilde{\mathbf{S}}_i^{co}, \nu + n_i^o + 2)}{T_{n_i}(\mathbf{V}_i | \boldsymbol{\mu}_i^{co}, \mathbf{S}_i^{co}, \nu + n_i^o)}$$

where $\widetilde{\mathbf{S}}_i^{co} = \left(\frac{\nu + Q(\mathbf{y}_i^o)}{\nu + 2 + n_i^o} \right) \boldsymbol{\Sigma}_i^{cc,o}$, $\widehat{\mathbf{w}}_i^c = E[\mathbf{W}_i]$, and $\widehat{\mathbf{w}}_i^{2c} = E[\mathbf{W}_i \mathbf{W}_i^\top]$, with $\mathbf{W}_i \sim T_{n_i^c}(\boldsymbol{\mu}_i^{co}, \widetilde{\mathbf{S}}_i^{co}, \nu + n_i^o + 2; \mathbb{A}_i^c)$ and $\boldsymbol{\Sigma}_i^{cc,o}$, $\boldsymbol{\mu}_i^{co}$, and $\mathbf{S}_i^{co}$ are as in equation (5).

As was mentioned in Proposition 4, formulas for $E[\mathbf{W}]$ and $E[\mathbf{W}\mathbf{W}^\top]$, where $\mathbf{W} \sim T_p(\boldsymbol{\mu}, \boldsymbol{\Sigma}, \nu; \mathbb{A})$, can be found in Ho et al.²³ For the computation of multivariate Student's t -cdf we used the *pmvt* function of the *mvtnorm* package³³ from R software. Additional details about the ECM algorithm for our proposed t -MLC model can be found in Appendix 1.

4.1 The expected information matrix

Louis³⁴ proposed a technique for computing the observed information matrix within the EM algorithm framework. Using this method, and from the results given by Lange et al.,²⁷ we can find an asymptotic approximation for the variances of the fixed effects in the t -MLC model. This approximation is given by

$$\mathbf{J}_{\beta\beta} = \text{Var}(\widehat{\boldsymbol{\beta}}) = \left(\sum_{i=1}^n \frac{\nu + n_i}{\nu + n_i + 2} \mathbf{X}_i^\top \boldsymbol{\Sigma}_i^{-1} \mathbf{X}_i - \sum_{i=1}^n \mathbf{X}_i^\top \boldsymbol{\Sigma}_i^{-1} (\mathbf{B}_i + \mathbf{D}_i) \boldsymbol{\Sigma}_i^{-1} \mathbf{X}_i \right)^{-1} \quad (10)$$

where $\mathbf{B}_i = \text{Var}\left[\frac{\nu+n_i}{\nu+Q(\mathbf{y}_i)}(\mathbf{y}_i - \mathbf{X}_i\boldsymbol{\beta})|\mathbf{V}_i, \mathbf{C}_i\right]$ and $\mathbf{D}_i = \left(\frac{2}{\nu+n_i}\right)(\widehat{u\mathbf{y}}_i^* - \widehat{u}_i^*\mathbf{X}_i\boldsymbol{\beta})(\widehat{u\mathbf{y}}_i^* - \widehat{u}_i^*\mathbf{X}_i\boldsymbol{\beta})^\top$, with $\mathbf{y}_i \sim Tt_{n_i}(\mathbf{X}_i\boldsymbol{\beta}, \boldsymbol{\Sigma}_i, \nu; \mathbb{A}_i)$, where $\mathbb{A}_i = \{\mathbf{y}_i \in \mathbb{R}^{n_i} | \mathbf{y}_i \leq \mathbf{a}_i\}$, with $\mathbf{y}_i = (y_{i1}, \dots, y_{in_i})^\top$ and $\mathbf{a}_i = (a_{i1}, \dots, a_{in_i})^\top$.

Note that clearly $\mathbf{B}_i$ and $\mathbf{D}_i$ depend on the following quantities

- $\widehat{u\mathbf{y}}_i^{2*} = E\left[\left(\frac{\nu+n_i}{\nu+Q(\mathbf{y}_i)}\right)^2 \mathbf{y}_i \mathbf{y}_i^\top | \mathbf{V}_i, \mathbf{C}_i, \widehat{\boldsymbol{\theta}}\right], \quad \widehat{u\mathbf{y}}_i^* = E\left[\left(\frac{\nu+n_i}{\nu+Q(\mathbf{y}_i)}\right)^2 \mathbf{y}_i | \mathbf{V}_i, \mathbf{C}_i, \widehat{\boldsymbol{\theta}}\right]$

and

- $\widehat{u}_i^* = E\left[\left(\frac{\nu+n_i}{\nu+Q(\mathbf{y}_i)}\right)^2 | \mathbf{V}_i, \mathbf{C}_i, \widehat{\boldsymbol{\theta}}\right]$

After some algebraic manipulations, we have three possible scenarios for the calculation of these quantities:

- If individual i has only noncensored components, then

$$\widehat{u\mathbf{y}}_i^{2*} = \left(\frac{\nu+n_i}{\nu+Q(\mathbf{y}_i)}\right)^2 \mathbf{y}_i \mathbf{y}_i^\top, \quad \widehat{u\mathbf{y}}_i^* = \left(\frac{\nu+n_i}{\nu+Q(\mathbf{y}_i)}\right)^2 \mathbf{y}_i, \quad \widehat{u}_i^* = \left(\frac{\nu+n_i}{\nu+Q(\mathbf{y}_i)}\right)^2$$

where $Q(\mathbf{y}_i) = (\mathbf{y}_i - \mathbf{X}_i\boldsymbol{\beta})^\top \boldsymbol{\Sigma}_i^{-1} (\mathbf{y}_i - \mathbf{X}_i\boldsymbol{\beta})$.

- If individual i has only censored components then from Proposition 2

$$\begin{aligned} \widehat{u\mathbf{y}}_i^{2*} &= c_{n_i}(\nu, 2) \frac{T_{n_i}(\mathbf{V}_i | \widehat{\boldsymbol{\mu}}_i, \widehat{\boldsymbol{\Sigma}}_i^*, \nu+4)}{T_{n_i}(\mathbf{V}_i | \widehat{\boldsymbol{\mu}}_i, \widehat{\boldsymbol{\Sigma}}_i, \nu)} E[\mathbf{W}_i \mathbf{W}_i^\top], \\ \widehat{u\mathbf{y}}_i^* &= c_{n_i}(\nu, 2) \frac{T_{n_i}(\mathbf{V}_i | \widehat{\boldsymbol{\mu}}_i, \widehat{\boldsymbol{\Sigma}}_i^*, \nu+4)}{T_{n_i}(\mathbf{V}_i | \widehat{\boldsymbol{\mu}}_i, \widehat{\boldsymbol{\Sigma}}_i, \nu)} E[\mathbf{W}_i], \\ \widehat{u}_i^* &= c_{n_i}(\nu, 2) \frac{T_{n_i}(\mathbf{V}_i | \widehat{\boldsymbol{\mu}}_i, \widehat{\boldsymbol{\Sigma}}_i^*, \nu+4)}{T_{n_i}(\mathbf{V}_i | \widehat{\boldsymbol{\mu}}_i, \widehat{\boldsymbol{\Sigma}}_i, \nu)} \end{aligned}$$

where $\mathbf{W}_i \sim Tt_{n_i}(\widehat{\boldsymbol{\mu}}_i, \widehat{\boldsymbol{\Sigma}}_i, \nu+4, \mathbb{A}_i)$, $\widehat{\boldsymbol{\mu}}_i = \mathbf{X}_i\boldsymbol{\beta}$, $\widehat{\boldsymbol{\Sigma}}_i^* = \frac{\nu}{\nu+4} \widehat{\boldsymbol{\Sigma}}_i$ and

$$c_{n_i}(\nu, 2) = \left(\frac{\nu+n_i}{\nu}\right)^2 \left[\frac{\Gamma(\frac{\nu+n_i}{2}) \Gamma(\frac{\nu+4}{2})}{\Gamma(\frac{\nu}{2}) \Gamma(\frac{\nu+n_i+4}{2})} \right]$$

- If individual i has censored and uncensored components, then from Proposition 3 and the fact that $\{\mathbf{y}_i | \mathbf{V}_i, \mathbf{C}_i\}$, $\{\mathbf{y}_i^o | \mathbf{V}_i, \mathbf{C}_i, \mathbf{y}_i^o\}$ and $\{\mathbf{y}_i^c | \mathbf{V}_i, \mathbf{C}_i, \mathbf{y}_i^o\}$ are equivalent processes, we have

$$\widehat{u\mathbf{y}}_i^{2*} = \begin{pmatrix} \widehat{u}_i^* \mathbf{y}_i^o \mathbf{y}_i^{o\top} & \widehat{u}_i^* \mathbf{y}_i^o \widehat{\mathbf{w}}_i^{c\top} \\ \widehat{u}_i^* \widehat{\mathbf{w}}_i^c \mathbf{y}_i^{o\top} & \widehat{u}_i^* \widehat{\mathbf{w}}_i^c \widehat{\mathbf{w}}_i^{c\top} \end{pmatrix}, \quad \widehat{u\mathbf{y}}_i^* = \text{vec}(\widehat{u}_i^* \mathbf{y}_i^o, \widehat{\mathbf{w}}_i^c),$$

$$\widehat{u}_i^* = \left(\frac{d_{n_i}}{(v + Q(\mathbf{y}_i^o))^2} \right) \frac{T_{n_i^c}(\mathbf{V}_i | \boldsymbol{\mu}_i^{co}, \widetilde{\mathbf{S}}_i^{co}, v + n_i^o + 4)}{T_{n_i^c}(\mathbf{V}_i | \boldsymbol{\mu}_i^{co}, \boldsymbol{\Sigma}_i^{co}, v + n_i^o)}$$

where

$$d_{n_i} = (v + n_i)^2 \left(\frac{\Gamma(\frac{n_i+v}{2}) \Gamma(\frac{n_i^o+v+4}{2})}{\Gamma(\frac{n_i^o+v}{2}) \Gamma(\frac{n_i+v+4}{2})} \right)$$

- $\widetilde{\mathbf{S}}_i^{co} = \left(\frac{v+Q(\mathbf{y}_i^o)}{v+4+n_i^o} \right) \widehat{\boldsymbol{\Sigma}}_i^{cc.o}$, $\widehat{\mathbf{W}}_i^c = E[\mathbf{W}_i]$, and $\widehat{\mathbf{w}}_i^c \widehat{\mathbf{w}}_i^{c\top} = E[\mathbf{W}_i \mathbf{W}_i^\top]$, with $\mathbf{W}_i \sim T_{n_i^c}(\boldsymbol{\mu}_i^{co}, \widetilde{\mathbf{S}}_i^{co}, v + n_i^o + 2, \mathbb{A}_i^c)$ and $\boldsymbol{\mu}_i^{co}$, $\boldsymbol{\Sigma}_i^{cc.o}$, and $\mathbf{S}_i^{co}$ as defined in Subsection 3.3.

Asymptotic confidence intervals for the fixed effects and hypothesis tests as well are obtained assuming that the ML estimates $\widehat{\boldsymbol{\beta}}$ have approximately a $N_p(\boldsymbol{\beta}, \mathbf{J}_{\beta\beta}^{-1})$ distribution. In practice, $\mathbf{J}_{\beta\beta}$ is usually unknown and it needs to be replaced by its ML estimates $\widehat{\mathbf{J}}_{\beta\beta}$.

4.2 Imputation of censored components

Let $\mathbf{y}_i^{(c)}$ be the true unobserved response vector for the censored components of the i th subject. Now, as a by-product of the EM algorithm we can obtain the predictor of the censored components, denoted by $\widetilde{\mathbf{y}}_i^{(c)}$, as follows

$$\widetilde{\mathbf{y}}_i^{(c)} = E[\mathbf{y}_i | \mathbf{y}_i^o, \mathbf{V}_i, \mathbf{C}_i, \widehat{\boldsymbol{\theta}}] \quad (11)$$

which is obtained considering two possible cases:

- (1) If subject i has only censored components

$$\widetilde{\mathbf{y}}_i^{(c)} = E[\mathbf{y}_i | \mathbf{V}_i, \mathbf{C}_i, \widehat{\boldsymbol{\theta}}]$$

where $\mathbf{y}_i | \mathbf{V}_i, \mathbf{C}_i, \widehat{\boldsymbol{\theta}} \sim T_{n_i}(\mathbf{X}_i \widehat{\boldsymbol{\beta}}, \widehat{\boldsymbol{\Sigma}}_i, v; \mathbb{A}_i)$, with $\mathbb{A}_i = \{\mathbf{y}_i \in \mathbb{R}^{n_i} | \mathbf{y}_i \leq \mathbf{a}_i\}$, $\mathbf{y}_i = (y_{i1}, \dots, y_{ini})^\top$ and $\mathbf{a}_i = (a_{i1}, \dots, a_{ini})^\top$. This expression is obtained using Proposition 4.

- (2) If subject i has censored and uncensored components, then from Proposition 3 with $r=0$ and $k=1$, we have that

$$\widetilde{\mathbf{y}}_i^{(c)} = E[\mathbf{y}_i^c | \mathbf{y}_i^o, \mathbf{V}_i, \mathbf{C}_i, \widehat{\boldsymbol{\theta}}]$$

with $\mathbf{y}_i^c | \mathbf{y}_i^o \sim T_{n_i^c}(\widehat{\boldsymbol{\mu}}_i^{co}, \widehat{\mathbf{S}}_i^{co}, v + n_i^o; \mathbb{A}_i)$ where $\mathbb{A}_i = \{\mathbf{y}_i \in \mathbb{R}^{n_i} | \mathbf{y}_i \leq \mathbf{a}_i\}$ with $\mathbf{y}_i = (y_{i1}, \dots, y_{ini})^\top$, $\mathbf{a}_i = (a_{i1}, \dots, a_{ini})^\top$, and

$$\widehat{\boldsymbol{\mu}}_i^{co} = \mathbf{X}_i^c \widehat{\boldsymbol{\beta}} + \widehat{\boldsymbol{\Sigma}}_i^{co} \widehat{\boldsymbol{\Sigma}}_i^{oo-1} (\mathbf{y}_i^o - \mathbf{X}_i^o \widehat{\boldsymbol{\beta}}), \quad \widehat{\mathbf{S}}_i^{co} = \left(\frac{v + Q(\mathbf{y}_i^o)}{v + n_i^o} \right) \widehat{\boldsymbol{\Sigma}}_i^{cc.o}$$

with $\widehat{\boldsymbol{\Sigma}}_i^{cc.o} = \widehat{\boldsymbol{\Sigma}}_i^{cc} - \widehat{\boldsymbol{\Sigma}}_i^{co} \widehat{\boldsymbol{\Sigma}}_i^{oo-1} \widehat{\boldsymbol{\Sigma}}_i^{oc}$ and $Q(\mathbf{y}_i^o) = (\mathbf{y}_i^o - \mathbf{X}_i^o \widehat{\boldsymbol{\beta}})^\top \widehat{\boldsymbol{\Sigma}}_i^{oo-1} (\mathbf{y}_i^o - \mathbf{X}_i^o \widehat{\boldsymbol{\beta}})$.

5 Prediction of future values

The problem related to the prediction of future values has a great impact in many practical applications. Rao³⁵ pointed out that the predictive accuracy of future observations can be taken as an alternative measure of “goodness of fit.” In order to propose a strategy for generating predicted values from our t -MLC model, we used the approach proposed by Wang.¹⁷ Thus, let $\mathbf{y}_{i,obs}$ be an observed response vector of dimension $n_{i,obs} \times 1$ for a new subject i over the first portion of time and $\mathbf{y}_{i,pred}$ the corresponding $n_{i,pred} \times 1$ response vector over the future portion of time. Let $\bar{\mathbf{X}}_i = (\mathbf{X}_{i,obs}, \mathbf{X}_{i,pred})$ be the $(n_{i,obs} + n_{i,pred}) \times p$ design matrix corresponding to $\bar{\mathbf{y}}_i = (\mathbf{y}_{i,obs}^\top, \mathbf{y}_{i,pred}^\top)^\top$.

To deal with the censored values existing in $\mathbf{y}_{i,obs}$, we used the imputation procedure presented in Subsection 4.2. Therefore, when the censored values are imputed, a complete dataset, denoted by $\mathbf{y}_{i,obs^*}$, is obtained. The reason to use the imputation procedure is that we avoid to compute truncated conditional expectations of multivariate Student’s t -distribution originated by the censoring scheme. Hence, we have that

$$\bar{\mathbf{y}}_i^* = (\mathbf{y}_{i,obs^*}^\top, \mathbf{y}_{i,pred}^\top)^\top \sim t_{n_{i,obs}+n_{i,pred}}(\mathbf{X}_i\boldsymbol{\beta}, \boldsymbol{\Sigma}_i, \nu)$$

where the matrix $\boldsymbol{\Sigma}_i$ defined in equation (3) can be represented by

$$\boldsymbol{\Sigma}_i = \begin{pmatrix} \boldsymbol{\Sigma}_i^{obs^*,obs^*} & \boldsymbol{\Sigma}_i^{obs^*,pred} \\ \boldsymbol{\Sigma}_i^{pred,obs^*} & \boldsymbol{\Sigma}_i^{pred,pred} \end{pmatrix}.$$

As was mentioned in Wang¹⁷ and Rao,³⁶ the best linear predictor of $\mathbf{y}_{i,pred}$ with respect to the minimum mean squared error (MSE) criterion is the conditional expectation of $\mathbf{y}_{i,pred}$ given $\mathbf{y}_{i,obs^*}$, which, from Proposition 1, is given by

$$\hat{\mathbf{y}}_{i,pred}(\boldsymbol{\theta}) = \mathbf{X}_{i,pred}\boldsymbol{\beta} + \boldsymbol{\Sigma}_i^{pred,obs^*}\boldsymbol{\Sigma}_i^{obs^*,obs^*}{}^{-1}(\mathbf{y}_{i,obs^*} - \mathbf{X}_{i,obs^*}\boldsymbol{\beta}) \quad (12)$$

Therefore, $\mathbf{y}_{i,pred}$ can be estimated directly by substituting $\hat{\boldsymbol{\theta}}$ into equation (12), leading to $\widehat{\mathbf{y}}_{i,pred} = \hat{\mathbf{y}}_{i,pred}(\hat{\boldsymbol{\theta}})$.

6 Simulation studies

In order to study the performance of the proposed method, we present two simulations studies. The first one shows the performance of t -MLC with DEC structure on the imputation procedure. The second one shows the asymptotic behavior of the ML estimates for our proposed model.

For both simulation schemes, we considered the t -MLC model defined in Subsection 3.2, with parameters setting at $\beta_1=2.5$, $\beta_2=4$, $\sigma^2=4$, $\phi_1=0.8$, and $\phi_2 = 1$. In addition, the time points are set as $\mathbf{t}_i = (1, 3, 5, 7, 10, 14)^\top$, for $i = 1, \dots, n$.

6.1 Imputation performance

As was mentioned earlier, the goal of this simulation study is to compare the performance of the t -MLC models with DEC structure under two scenarios: (a) when the parameters ϕ_1 and ϕ_2 are unknown and estimated from the data, called unspecified (U) structure, and (b) when $\mathbf{E}_i = \mathbf{I}_{n_i}$

with $\mathbf{I}_{n_i}$ representing the identity matrix of order $n_i \times n_i$, that is, when an uncorrelated (UNC) structure is considered. For these purposes, we proceeded as follows:

- (1) We generated $M = 100$ datasets of size $n = 300$ from the t -MLC model with a DEC structure $\mathbf{E}_i = 0.8^{|t_{ij} - t_{ik}|}$, under four different settings of censoring proportions say, $\gamma = 5, 15, 25$, and 35% . It is important to note that the goal here is to study the effect of the level of censoring in the estimation under misspecification of the correlation structure.
- (2) All the censored observations were imputed using the mechanism described in Subsection 4.2 and considering both U and UNC structure.

In order to compare performance of the U and UNC structures through the EM imputation defined in equation (11), we utilized two empirical discrepancy measures called the mean absolute error (MAE) and mean square error (MSE).^{17,37} They are defined by

$$\text{MAE} = \frac{1}{k} \sum_{i,j} |y_{ij} - \tilde{y}_{ij}| \quad \text{and} \quad \text{MSE} = \frac{1}{k} \sum_{i,j} (y_{ij} - \tilde{y}_{ij})^2 \quad (13)$$

where y_{ij} is the original simulated value (before being considered as a censored observation) and $\tilde{y}_{ij}$ is the EM imputation, for $i = 1, \dots, 300$ and $j = 1, \dots, 6$. Note that for $\gamma = 5\%$ we have that $k = 90$, for $\gamma = 15\%$, $k = 270$, for $\gamma = 25\%$, $k = 450$, and for $\gamma = 35\%$, $k = 630$.

Arithmetic means of MSE and MAE over the 100 datasets are displayed in Table 2 and Figure 2. We can see that in all cases, the U structure presents the smallest MSE and MAE than the UNC one, as expected.

6.2 Asymptotic properties

In this simulation study, we analyze the absolute bias (Bias) and MSE of the regression coefficient estimates obtained from the t -MLC model for six different sample sizes ($n = 50, 100, 200, 300, 400$, and 600). These measures are defined by

$$\text{Bias}(\theta_i) = \frac{1}{M} \sum_{j=1}^M \left| \hat{\theta}_i^{(j)} - \theta_i \right| \quad \text{and} \quad \text{MSE}(\theta_i) = \frac{1}{M} \sum_{j=1}^M \left(\hat{\theta}_i^{(j)} - \theta_i \right)^2$$

where $\hat{\theta}_i^{(j)}$ is the ML estimate of the parameter θ_i for the j th sample.

Table 2. Simulated data. Arithmetic means of the MAE and MSE over $M = 100$ datasets.

Censoring level (%)	Correlation structure			
	Unspecified U		Uncorrelated UNC	
	MAE	MSE	MAE	MSE
5	1.120052	2.744973	1.199131	3.075949
15	1.293753	3.106423	1.563213	4.340442
25	1.409025	3.902546	1.684068	5.475168
35	1.568360	4.647703	1.830202	6.170776

The idea of this simulation is to provide empirical evidence about consistency of the ML estimators under the t -MLC model with DEC structure. For each sample size, we generate $M=100$ dataset with 5% of censoring proportion. In this simulation scheme the parameter ϕ_2 is fixed at 1 and thus a continuous-time AR(1) model is considered. Using the ECM algorithm, the absolute bias and MSE for each parameter over the 100 datasets were computed.

From Figure 3 we can see that the absolute bias and the MSE tend to zero as the sample size increases. As is expected, under the t -MLC model the EM algorithm provides estimates with good asymptotic properties.

7 Application

We applied our method to the UTI data described in Section 2. This dataset consists of 362 observations, 26 were below the detection limits (50 or 400 copies/ml). The UTI data were analyzed previously by Lachos et al.,¹³ where it was observed that inferences based on Gaussian assumptions are questionable. Consequently, we revisited this dataset with the aim of carrying out robust inference by considering the Student's t model. We consider our proposed t -MLC model with the DEC correlation structure $\Sigma_i = \sigma^2 \mathbf{E}_i$ defined in Subsection 3.2 and for the sake of model comparison, we also fit the normal MLC (N-MLC) counterparts, which can be treated as the reduced t -MLC as ν tends to infinity. Here we have that y_{ij} is the $\log_{10}$ HIV-1 RNA for subject i at time t_j , with $t_1=0$, $t_2=1$, $t_3=3$, $t_4=6$, $t_5=9$, $t_6=12$, $t_7=18$, and $t_8=24$.

It is important to stress that the t -LMM model related to the UTI data, mentioned in the introduction, corresponds to a LMM with a random intercept. In that model, Matos et al.²² consider that the random effect follows a Student's t -distribution. It is well known that this type of model may not adequately account for the serial correlation generated by the longitudinal data arising from medical studies such as HIV studies, since the correlation structure induced by the mixed model with random intercept generates a model with compound symmetric correlation structure. Moreover, if we misspecified the distributional assumption for the random effects, we may severely affect the likelihood-based inference and predictions as well McCulloch and

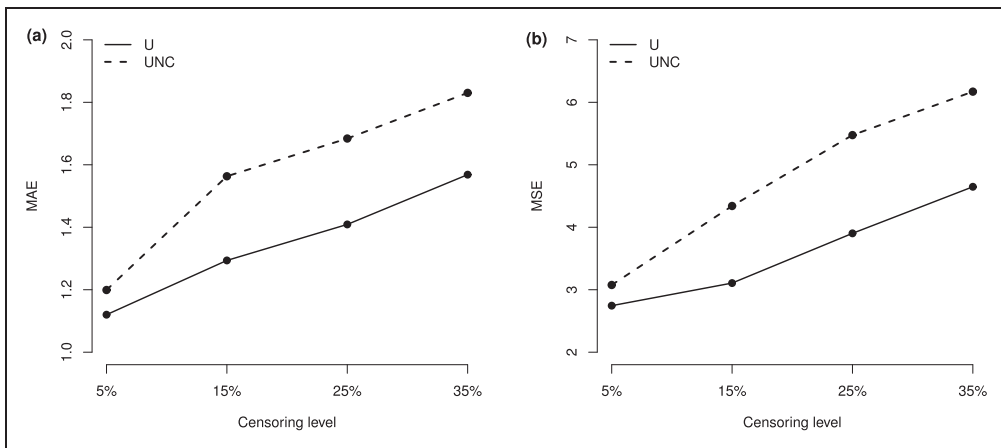


Figure 2. Simulated data. Arithmetic means of (a) MAE and (b) MSE over $M = 100$ datasets under the t -MLC model with U and UNC structures.

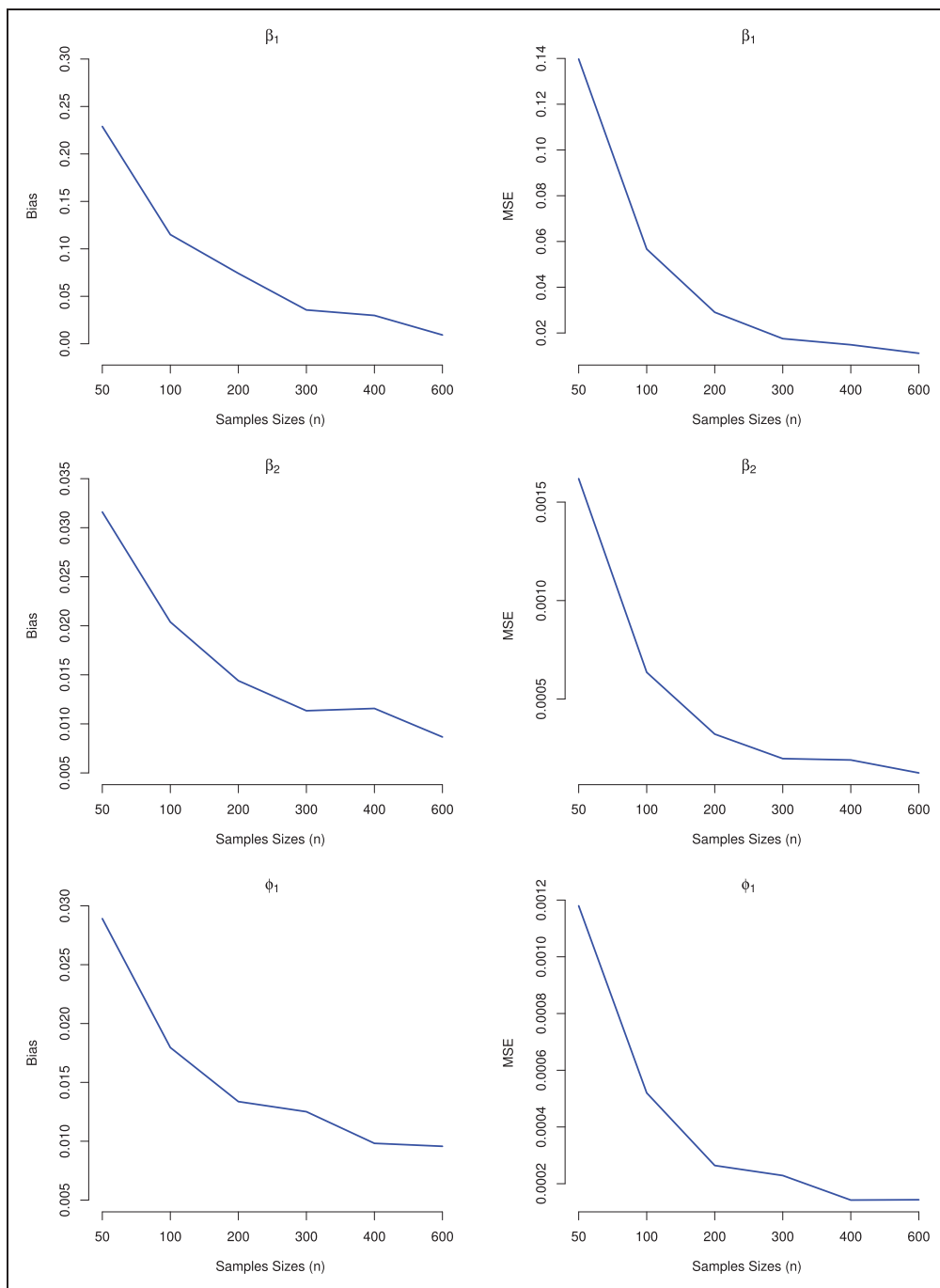


Figure 3. Simulated data. Bias (first column) and MSE (second column) of parameter estimates in the t -MLC model under 5% of censoring.

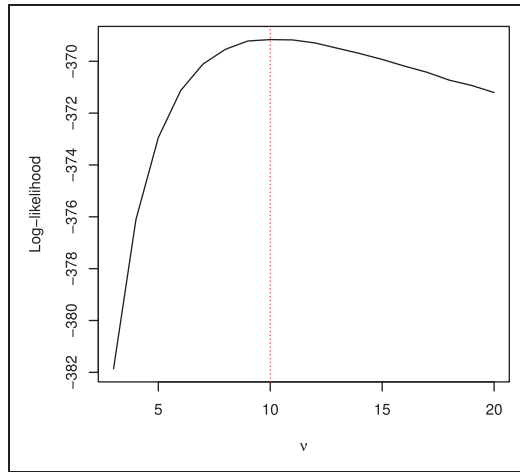


Figure 4. UTI data. Plot of the profile log-likelihood of the degrees of freedom ν .

Neuhaus.³⁸ For those reasons, our approach do not consider the use of random effects. Instead, we consider the DEC structure to deal with the correlation generated by the HIV data. Moreover, using the DEC structure we can also obtain the compound symmetry structure ($\phi_2 = 0$) as a particular case, which is equivalent to the model considered by Matos et al.²² From our point of view, the DEC structure is a flexible and parsimonious way to deal with correlation in longitudinal data. It avoids specifying the probability distribution for the random effects and allows us to obtain different correlation structures.

We consider four cases of correlation structure induced by the specification of the matrix $\mathbf{E}_i$, namely (a) the uncorrelated (UNC) structure, (b) the continuous-time AR(1) structure, (c) the MA(1) structure, and (d) the unknown (U) structure (when ϕ_1 and ϕ_2 are unknown).

The degrees of freedom parameter ν was fixed at the value that maximizes the t -MLC likelihood function. Figure 4 shows that the likelihood function reaches the maximum at $\nu = 10$, indicating the lack of adequacy of the normal assumption for the UTI data. The ML estimates of the other parameters were obtained using the ECM algorithm described in Section 4, with starting values obtained through the library *lme4*.²⁵

Table 3 presents the ML estimates and standard errors of the regression parameters β for the t -MLC and N-MLC models. Although the estimates are quite similar in both cases, the standard errors are in general smaller under the Student's t , indicating that our proposed censored model (t -LMC) produces more precise estimates.

Note that we fitted eight models, resulting from the combinations of the four correlations structures, say UNC, AR(1), MA(1), and U, and two distributional assumptions (normal and Student's t). The value of the log-likelihood function and the AIC and BIC criteria for these eight models are presented in Table 4, where we can see that the t -MLC outperforms consistently the normal counterpart in all cases. In particular, these criteria indicate a preference of the unspecified correlation structure (U structure), that is, when the parameters ϕ_1 and ϕ_2 of the matrix $\mathbf{E}_i$ are estimated from the data.

The regression coefficients β_j , for $j = 1, \dots, 8$, increase gradually under the two models. This is the evidence of the negative effect of the antiretroviral therapy interruption on the viral load levels. In other words, the viral load increments consistently along the time when the antiretroviral therapy

Table 3. UTI data. ML estimation (Est) and standard errors (SE) for the regression coefficients under the normal and t-MLC models with different DEC structures.

Parameters	N-MLC							
	UNC		AR(1)		MA(1)		U	
	Est	SE	Est	SE	Est	SE	Est	SE
β_1	3.6160	0.0153	3.6334	0.0162	3.6194	0.0150	3.6196	0.0156
β_2	4.1527	0.0172	4.2095	0.0168	4.1825	0.0166	4.1834	0.0164
β_3	4.2381	0.0184	4.2502	0.0182	4.2384	0.0181	4.2568	0.0169
β_4	4.3727	0.0187	4.3224	0.0189	4.3729	0.0184	4.3738	0.0170
β_5	4.3650	0.0248	4.4680	0.0237	4.3652	0.0245	4.5791	0.0195
β_6	4.2326	0.0313	4.3781	0.0303	4.2327	0.0309	4.5819	0.0221
β_7	4.3258	0.0444	4.3749	0.0463	4.3260	0.0438	4.6879	0.0275
β_8	4.5620	0.0818	4.5762	0.0842	4.5620	0.0807	4.8061	0.0418
σ^2	1.0631		1.1498		1.0486		1.1053	
ϕ_1	—		0.8251		0.4068		0.7027	
ϕ_2	—		1.00		∞		0.0286	
Parameters	t-MLC							
	UNC		AR(1)		MA(1)		U	
	Est	SE	Est	SE	Est	SE	Est	SE
β_1	3.6511	0.0120	3.6410	0.0155	3.6578	0.0120	3.6330	0.0153
β_2	4.2386	0.0146	4.3022	0.0172	4.2706	0.0144	4.2697	0.0171
β_3	4.3149	0.0156	4.3312	0.0187	4.3246	0.0156	4.3290	0.0177
β_4	4.4715	0.0159	4.4297	0.0195	4.4792	0.0159	4.4715	0.0178
β_5	4.5268	0.0210	4.5476	0.0248	4.5293	0.0209	4.6359	0.0206
β_6	4.3923	0.0267	4.4435	0.0317	4.3963	0.0266	4.6238	0.0235
β_7	4.5012	0.0373	4.4660	0.0475	4.5092	0.0377	4.7082	0.0295
β_8	4.6896	0.0692	4.6481	0.0863	4.5092	0.0687	4.7998	0.0455
σ^2	0.8092		1.0272		0.8003		1.0103	
ϕ_1	—		0.7754		0.2752		0.6629	
ϕ_2	—		1.00		∞		0.0222	
ν	10.00		10.00		10.00		10.00	—

Table 4. UTI data. Comparison between the normal and t-MLC models using different model selection criteria.

Criteria	N-MLC				t-MLC			
	UNC	AR(1)	MA(1)	U	UNC	AR(1)	MA(1)	U
Log-likelihood	−524.166	−463.043	−516.507	−411.926	−484.165	−421.249	−476.647	−369.129
AIC	1066.333	946.087	1053.014	845.852	986.331	862.498	973.295	760.259
BIC	1101.357	985.004	1091.931	888.660	1021.357	901.415	1012.212	803.067
AIC_{corr}	1066.844	946.714	1053.641	846.607	986.843	863.125	973.922	761.014

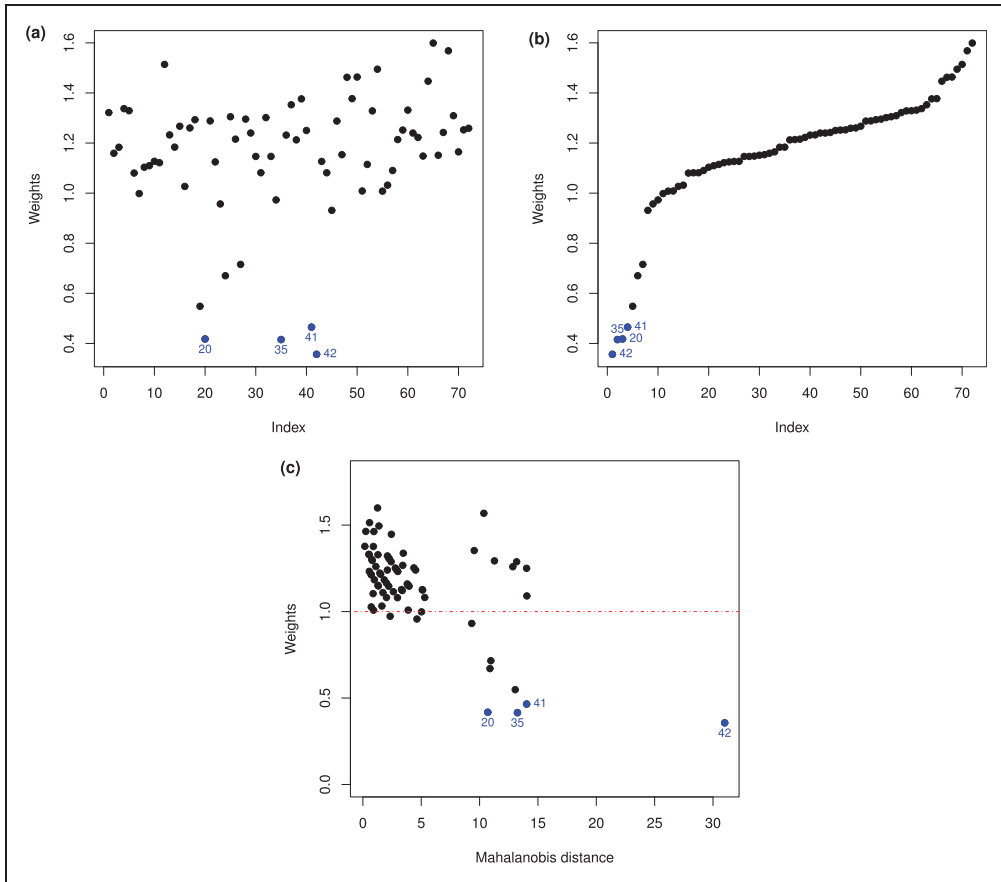


Figure 5. UTI data. Estimated weights $\hat{u}_i$ for the t -MLC model under U structure.

Table 5. UTI data. Evaluation of the prediction accuracy for the t -MLC model with different DEC structures.

For individuals with $n_i \geq 5$ (41 individuals)								
Forecast	U		AR(I)		MA(I)		UNC	
	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE
One step	0.3218993	0.1768571	0.4968972	0.3456072	0.7392737	0.8093977	0.7239416	0.8128087
Two step	0.4143890	0.2827286	0.5987888	0.6146529	0.7311850	0.9008162	0.7240895	0.9038104
For individuals with $n_i \geq 6$ (29 individuals)								
Forecast	U		AR(I)		MA(I)		UNC	
	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE
One step	0.3308357	0.1912845	0.4388853	0.2702304	0.6237170	0.5197352	0.6030431	0.5267867
Two step	0.3721411	0.2159791	0.5222417	0.5049302	0.6417997	0.7027740	0.6307786	0.7051547

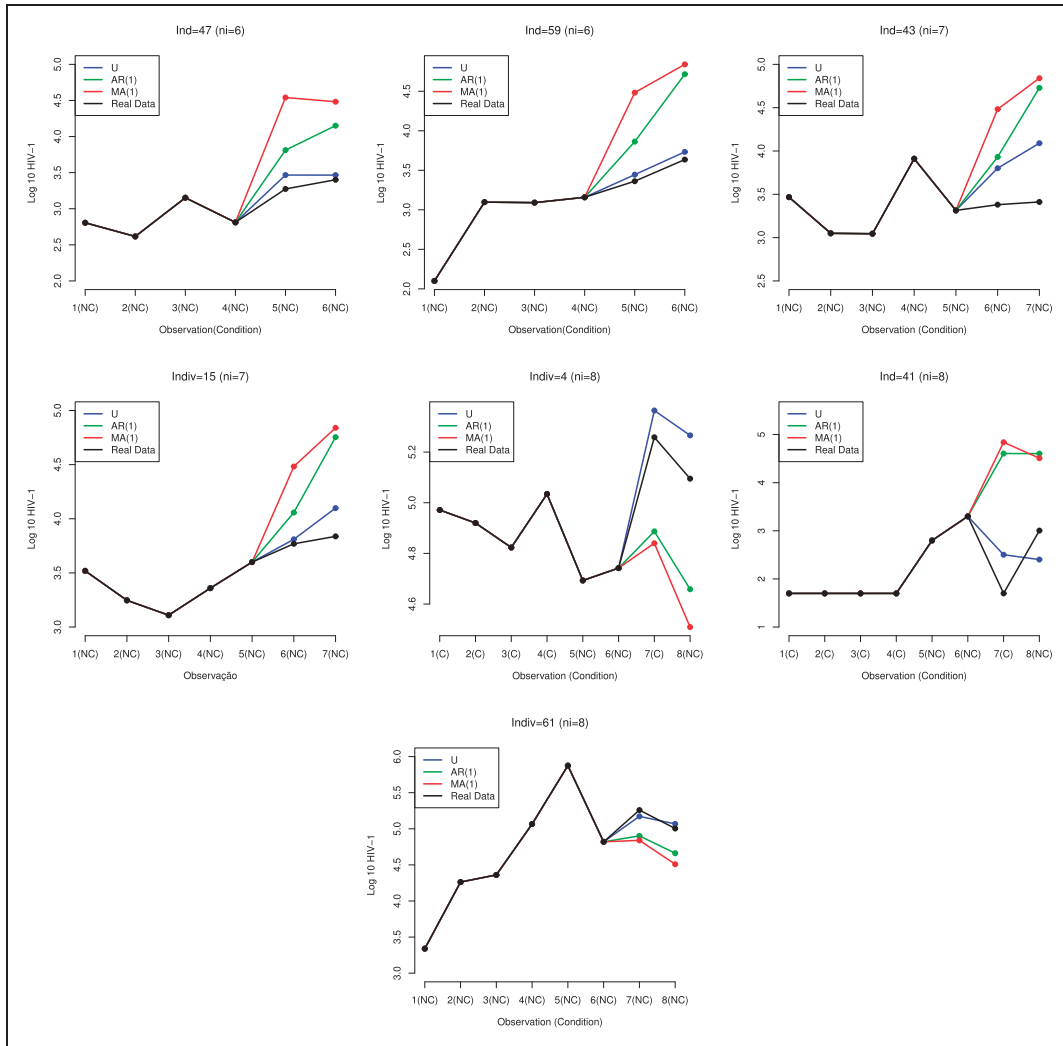


Figure 6. UTI data. Evaluation of the prediction performance for seven random subjects. The abbreviation “Ind” stands for “Individual.” n_i denotes the number of measurements. C and NC indicate if the measurement was censored or not, respectively.

begins to be interrupted. For the best model (t -MLC), the coefficients increase from 3.63 at the beginning of the study to 4.79 at the end of this. Note that considering an asymptotic 95% confidence interval, the estimates of all regression coefficients are significant. The estimate of the within-subject (σ^2) scale parameter (in $\log_{10}$ scale) is 1.01.

Outlying observations may affect the estimation of the parameters under assumptions of normality. Our t -MLC model with DEC structure accommodates these discrepant observations attributing to them small weights in the estimation procedure. The estimated weights ($\hat{u}_i, i = 1, \dots, 72$) for the t -MLC model with U structure are presented in Figure 5. In this figure, we observe that observations #20, #35, #41 and #42 present smaller weights, verifying the robust

aspects of the ML estimation under the Student's t -distribution. These results agree with those obtained by Lachos et al.¹³ under the Bayesian paradigm.

Now we turn our attention to the one-step-ahead and two-step-ahead forecast of future observations using the approach proposed in Section 5 for the UTI data.

As a simple illustration, we considered in the analysis the cases which were measured on at least five (41 individuals or 56%) and six occasions (29 individuals in total) and we predicted the last two measures. As in the simulation scheme presented in Subsection 6.1, we considered the MAE and MSE measures for comparing the performance of the prediction under different DEC structures. Table 5 presents the comparison between the predicted values (one-step-ahead and two-step-ahead) with the real ones, under the t -MLC model considering four different DEC structures, say, AR(1), MA(1), UNC, and U. Figure 6 shows the performance of prediction for individuals with different types of trajectories under three different correlation structures, namely the MA(1) (red line), AR(1) (green line), and U (blue line) for the individuals: #4, #15, #41, #43, #43, #47, #47 and #61. We can see from these results once again how the U structure outperforms the other correlation structures from a predictive point of view, i.e. the U structure generates predictive values close to the real ones.

Moreover, the two-step-ahead prediction performance of the N-MLC under the U structure provides values of the MAE and MSE equal to 0.3786 and 0.2321, respectively. Thus, from Table 5 we can conclude that the t -MLC model generates better predictive results under the U structure than the normal one.

8 Conclusions

We have proposed a robust approach to linear regression models with censored observations based on the multivariate Student's t -distribution, called the t -MLC model. This offers a flexible alternative for dealing with longitudinal censored data in the presence of outliers and/or influential observations. For modeling the autocorrelation existing among irregularly observed measures, a damped exponential correlation structure was adopted as proposed by Muñoz et al.¹⁸ A novel ECM algorithm to obtain the ML estimates is developed by exploring the statistical properties of the multivariate truncated Student's t -distribution. Our proposed algorithm has a closed-form expression for the E-step, based on formulas for the mean and variance of the truncated Student's t -distribution. We applied our methods to a recent AIDS study (freely downloadable from R), concluding that when the antiretroviral therapy is interrupted, the HIV-1 RNA levels in blood increase consistently along the period of evaluation. We also perform two simulation studies, showing the superiority of t -MLC model on the provision of more adequate results when the available data have censored components. Furthermore, the simulation results demonstrate that our method gives very competitive performance in terms of imputation when a DEC structure is considered. From these results it is encouraging that the use of the t -MLC model with DEC structure offers a better fit, protection against outliers, and more precise inferences.

An established way to validate the model (2) is to use the bootstrap technique (see e.g. Efron and Tibshirani³⁹ or Chernick⁴⁰). Here, the appropriate way to use bootstrap will be to bootstrap observations (Y_i, X_i) , $i = 1, \dots, n$. This means drawing randomly with replacement from the set of indices $\{1, \dots, n\}$ and obtaining a "new" sample of the same size n as the original sample. Since the bootstrap sampling is with replacement, it is quite likely here to draw twice or more the observation with the same index i , where $1 \leq i \leq n$. For each of the new sample we can recalculate the estimates of the parameter β in the model. After creating B samples we can have a data-driven approximation of the sampling distribution of the estimate $\hat{\beta}$ and a better idea on variability of $\hat{\beta}$. This now can be

compared with the chosen p dimensional Student's t -distribution to check whether this parametric model is reflecting data-driven sampling distribution of $\hat{\beta}$ and its variance.

Although the t -MLC considered here has shown great flexibility for modeling symmetric data, its robustness against outliers can be seriously affected by the presence of skewness. Recently, Lachos et al.⁴¹ (see also Bandyopadhyay et al.¹⁴) proposed a remedy to accommodate skewness and heavy-tailedness simultaneously, using scale mixtures of skew-normal distributions. We conjecture that our methods can be used under MLC models and should yield satisfactory results at the expense of additional complexity in the implementation. An in-depth investigation of such extensions is beyond the scope of the present paper, but it is an interesting topic for further research.

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Notes

1. Gamma (a, b) denotes a gamma distribution with a/b mean.

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Appendix I: Details of the ECM algorithm

In this appendix, we derive the ECM algorithm equations (7) to (9) for the t -MLC model. Let $\mathbf{y} = (\mathbf{y}_1^\top, \dots, \mathbf{y}_n^\top)^\top$, $\mathbf{u} = (u_1, \dots, u_n)^\top$, $\mathbf{V} = \text{vec}(\mathbf{V}_1, \dots, \mathbf{V}_n)$, and $\mathbf{C} = \text{vec}(\mathbf{C}_1, \dots, \mathbf{C}_n)$ such that we observe $(\mathbf{V}_i, \mathbf{C}_i)$ for the i -th subject. Treating $\mathbf{u}$ and $\mathbf{y}$ as hypothetical missing data, and augmenting with the observed data $\mathbf{V}$, $\mathbf{C}$, we set $\mathbf{y}_c = (\mathbf{C}^\top, \mathbf{V}^\top, \mathbf{y}^\top, \mathbf{u}^\top)^\top$.

Denoting the complete-data likelihood by $L(\cdot|\mathbf{C}^\top, \mathbf{V}^\top, \mathbf{y}^\top, \mathbf{u}^\top)$ and pdfs in general by $f(\cdot)$, we have that for $\boldsymbol{\theta} = (\boldsymbol{\beta}^\top, \sigma^2, \phi_1, \phi_2)^\top$

$$\begin{aligned} L(\boldsymbol{\theta}|\mathbf{C}^\top, \mathbf{V}^\top, \mathbf{y}^\top, \mathbf{u}^\top) &= f(\mathbf{y}|\mathbf{V}, \mathbf{C}, \mathbf{u})f(\mathbf{u}) \\ &= f(\mathbf{y}|\mathbf{u})f(\mathbf{u}) = \prod_{i=1}^n f(\mathbf{y}_i|u_i)h(u_i|v) \end{aligned}$$

Dropping unimportant constants, the complete-data log-likelihood function is given by

$$\begin{aligned}\ell_c(\boldsymbol{\theta}|\mathbf{y}_c) &= \log\{L[\boldsymbol{\theta}|\mathbf{y}_c]\} = \log\left\{\prod_{i=1}^n f(\mathbf{y}_i|u_i)h(u_i|\nu)\right\} \\ &= \sum_{i=1}^n \log\left\{(2\pi)^{-p/2}u_i^{1/2}|\Sigma_i|^{-1/2}\exp\left(\frac{u_i}{2}(\mathbf{y}_i - \mathbf{X}_i\boldsymbol{\beta})^\top \Sigma_i^{-1}(\mathbf{y}_i - \mathbf{X}_i\boldsymbol{\beta})\right)\right\} + \sum_{i=1}^n \log\{h(u_i|\nu)\}, \\ &= -\frac{1}{2}\sum_{i=1}^n \left[n_i \log \sigma^2 + \log |\mathbf{E}_i| + \frac{u_i}{\sigma^2}(\mathbf{y}_i - \mathbf{X}_i\boldsymbol{\beta})^\top \mathbf{E}_i^{-1}(\mathbf{y}_i - \mathbf{X}_i\boldsymbol{\beta})\right] + \sum_{i=1}^n \log\{h(u_i|\nu)\} + c\end{aligned}$$

where c is a constant that is independent of the parameter vector $\boldsymbol{\theta}$ and $h(u_i|\nu)$ is the gamma density (Gamma($\nu/2, \nu/2$)). Our EM-type algorithm (ECM) for the t -MLC model can be summarized in the following way.

E-step

Given the current value $\boldsymbol{\theta} = \widehat{\boldsymbol{\theta}}^{(k)}$, the E-step calculates the conditional expectation of the complete data log-likelihood function

$$\begin{aligned}Q(\boldsymbol{\theta}|\widehat{\boldsymbol{\theta}}^{(k)}) &= \sum_{i=1}^n Q_i(\boldsymbol{\theta}|\widehat{\boldsymbol{\theta}}^{(k)}), \\ &= \sum_{i=1}^n \left\{ -\frac{n_i}{2} \log \sigma^2 - \frac{1}{2} \log |\mathbf{E}_i| - \frac{1}{2\sigma^2} A_i^{(k)}(\boldsymbol{\beta}, \phi) \right\}\end{aligned}$$

with

$$A_i^{(k)}(\boldsymbol{\beta}, \phi) = \left[\text{tr}\left(u_i \widehat{\mathbf{y}}_i^{2(k)} \mathbf{E}_i^{-1}\right) - 2\boldsymbol{\beta} \mathbf{X}_i^\top \mathbf{E}_i^{-1} u_i \widehat{\mathbf{y}}_i^{(k)} + \hat{u}_i^{(k)} \boldsymbol{\beta} \mathbf{X}_i^\top \mathbf{E}_i^{-1} \mathbf{X}_i \boldsymbol{\beta} \right]$$

Note that in this case we do not consider the computation of $E[h(u_i|\nu)|\mathbf{V}, \mathbf{C}, \widehat{\boldsymbol{\theta}}^{(k)}]$ because ν is fixed.

CM step

The CM step then conditionally maximizes $Q(\boldsymbol{\theta}|\widehat{\boldsymbol{\theta}}^{(k)})$ with respect to $\boldsymbol{\theta} = (\boldsymbol{\beta}^\top, \sigma^2, \phi_1, \phi_2)^\top$ and obtains a new estimate $\widehat{\boldsymbol{\theta}}^{(k+1)}$

$$\begin{aligned}\frac{\partial Q(\boldsymbol{\theta}|\widehat{\boldsymbol{\theta}}^{(k)})}{\partial \boldsymbol{\beta}} &= \frac{1}{\sigma^2} \sum_{i=1}^n \left[\mathbf{X}_i^\top (\widehat{\mathbf{E}}_i^{(k)})^{-1} u_i \widehat{\mathbf{y}}_i^{(k)} - \left(\hat{u}_i^{(k)} \mathbf{X}_i^\top (\widehat{\mathbf{E}}_i^{(k)})^{-1} \mathbf{X}_i \right) \boldsymbol{\beta} \right]; \\ \frac{\partial Q(\boldsymbol{\theta}|\widehat{\boldsymbol{\theta}}^{(k)})}{\partial \sigma^2} &= -\frac{N}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^n A_i^{(k)}(\widehat{\boldsymbol{\beta}}^{(k+1)}, \widehat{\boldsymbol{\phi}}^{(k)})\end{aligned}$$

Thus, the solutions of $\frac{\partial Q(\theta|\theta^{(k)})}{\partial \beta} = 0$ and $\frac{\partial Q(\theta|\theta^{(k)})}{\partial \sigma^2} = 0$ are

$$\begin{aligned}\widehat{\beta}^{(k+1)} &= \left(\sum_{i=1}^n \widehat{u}_i^{(k)} \mathbf{X}_i^\top (\widehat{\mathbf{E}}_i^{(k)})^{-1} \mathbf{X}_i \right)^{-1} \sum_{i=1}^n \mathbf{X}_i^\top (\widehat{\mathbf{E}}_i^{(k)})^{-1} \widehat{u}_i^{(k)}, \\ \widehat{\sigma}^{2(k+1)} &= \frac{1}{N} \sum_{i=1}^n A_i^{(k)}(\widehat{\beta}^{(k+1)}, \widehat{\phi}^{(k)})\end{aligned}$$

where $N = \sum_{i=1}^n n_i$. We estimate ϕ by maximizing the marginal log-likelihood, circumventing the (in general) complicated task of computing $\frac{\partial \mathbf{E}_i}{\partial \phi}$. This strategy was used, for instance, by Wang¹⁷ and Wang and Fan.³⁷ Then

$$\widehat{\phi}^{(k+1)} = \underset{\phi}{\operatorname{argmax}} \left\{ -\frac{1}{2} \sum_{i=1}^n \left[\log(|\mathbf{E}_i|) + A_i^{(k)}(\widehat{\beta}^{(k+1)}, \phi) \right] \right\}$$

The algorithm is iterated until the distance involving two successive evaluations of the log-likelihood, $|\ell(\widehat{\theta}^{(k+1)})/\ell(\widehat{\theta}^{(k)}) - 1|$, is sufficiently small.